Jackie (Jingkai) Shao

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Work Experience

Manufacturing Controls Engineer **HMI Programming | Python | KUKA Robot Lauguage (KRL)**
OPmobility (Plastic Omnium) | Internship *April 2026 – May 2026*

- Developed and deployed KRL control systems on the KRC4 controller for filament winding applications on Type IV hydrogen mobility vessels, improving winding accuracy around the dome of the thermoplastic liner by 15%.
- Commissioned 3-axis Siemens SINAMICS S120 servo drive system via TIA Portal Startdrive, achieving ±5mm positional accuracy of the carriage guide and reducing delamination defects during the winding process.
- Developed an automation program using Python (pandas + NumPy) that parses ComposiCAD output files and cross-references with actual winding parameters from PLC Memory to flag deviations, cutting inspection time by 30%.

Mechanical Engineer **SolidWorks | AutoCAD | Triflex | Ansys | Welding**
Chart Industries, Inc. | Co-op *December 2025 – March 2026*

- Designed 10+ SolidWorks 3-D models using DFM/DFA principles and AutoCAD 2-D drawings (Part and Assembly) for the bulk storage tanks, improving the First Pass Yield (FPY) from 78% to 85%.
- Conducted a pipeline stress analysis using Triflex and added expansion loops or bellows (design for strength and stiffness) to prevent high concentrations of stress, achieving the $3.5 - 4.0$ safety factor outlined by ASME Section VIII.
- Conducted a Conjugate Heat Transfer Analysis in Ansys Fluent to simulate the heat transfer across the fluid boundary layers near the internal tank wall, resulting in a 4% decrease of Nusselt Number (Nu).
- Developed MIG/TIG welding procedures for external tank structures, redesigning joint geometries to improve weld accessibility and reduce weld time per assembly by 10%.

System Design & Automation Engineer **CATIA | Abaqus | KUKA.Sim | PLC Programming**
OPmobility (Plastic Omnium) | Internship *May 2025 – August 2025*

- Designed 5+ CATIA 3D models using DFM/DFA and 2D drawings (Part and Assembly) with GD&T for the thermoplastic liners, resulting in better compatibility with automated manufacturing.
- Conducted PLC programming (ladder logic) on the Siemens S7-1500 to reduce the wedge effect of the composite filament layers by developing a Tension Profile, improving manufacturing consistency from $C_{pk} < C_p$ to $Cpk \geq 1.33$.
- Performed non-linear FEA in Abaqus using raw data from proprietary ComposiCAD software to pinpoint structural weaknesses, predict burst pressure, and simulate pressure cycle testing, reducing prototyping iteration time by 30%.
- Developed a Digital Twin Robotic Arm in KUKA.Sim to perform a Gage R&R study on fibre winding angle, improving the measurement accuracy of the rotary encoders by 10%.
- Interfaced Siemens S7-1500 PLC with conveyors, KUKA robots, and Manufacturing Execution System (MES) over PROFINET, reducing inter-device communication faults by 10%.
- Implemented control reliable safety systems for safeguarding compliant with ANSI/RIA 15.06 and OSHA standards, integrating dual-channel E-stops, light curtains, and safety relays across 5+ machine cells.

Projects

Filament Winding Robotic Arm Digital Twin (KUKA.Sim) | *Github* **Associated with OPmobility**

- Reduced physical rework by 10+ hours by validating robot programs in KUKA.Sim digital twin prior to deployment, catching 5+ logic errors during pre-commissioning.

Cryogenics Systems Modeling | *Github* **Associated with Chart Industries**

- Designed a 3-D Modeling of LNG Cryogenic Storage Tank in SolidWorks with instrumentation panels (HMI + gauges) and piping functions (valves + regulators + reliefs), providing clear visualization to clients and technicians.

Education

Bachelor of Applied Science in Mechanical Engineering **September 2025 - June 2030**
University of Waterloo, Honours Program with Co-op | GPA: 3.7/4.0 *Ontario, Canada*

- **Relevant Coursework:** Engineering Design, C++ Programming, Material Science, Electrical Circuits, Mechanics, CFD/FEA Workshop
- **Student Activities:** System Design Engineering Researcher at the Waterloo Institute for Sustainable Aeronautics

Skills

Software: CATIA, SolidWorks, Abaqus, Ansys, AutoCAD, Triflex, Robotstudio, KUKA.Sim, Word, Excel, PowerPoint, Visio
Technical: Analog and Digital Circuits, Control Systems, Logic, Industrial Robots (Kuka, ABB), Servo Controllers (SINAMICS S210)
Safety: NFPA79/NEC/UL508A, ANSI/RIA15.06/OSHA, ASME BPVC Section XII, ASME Section VIII, ASME B31.3 (Process Piping)
Manufacturing: CNC Machining, Common Power and Hand Tools, Manual Lathe/Mill, Filament Winding, Blow/Injection Moulding, Welding
Programming: RAPID, KRL, C++, Python, SQL, MATLAB/Simulink, PLC Programming (Siemens), HMI/SCADA Programming